

MH370: combining impact and search uncertainty

We estimate where MH370 could have struck the ocean by simulating aircraft motion, comparing predicted satellite measurements with observations, and updating candidate probabilities. Satellite timing (BTO) constrains range; frequency shifts (BFO) constrain motion. The stored cruise ensemble began with 1.02 million candidates, allowing up to 16 direction, eight Mach and eight altitude command starts. For Bayesian prediction and updating, see Davey et al. (2016, Section 3.1). Fuel exhaustion is conditional on 00:15–00:19 UTC. Simulated unpowered continuations connect 00:11 states to computed impacts.

The two 00:19 BFOs may be unreliable. We compare four cases: neither, R600 only, R1200 only, or both usable without extra oscillator warm-up adjustments. All retain the corrected R600 BTO and an airborne transmitter through 00:19:37.443. Each separately fitted case receives an initial weight of 25%, before drift/search updates. This specifies a mixture of conditional estimates; it does not infer instrument reliability from the BFO fit scores.

Let $q_m(x)$ be the normalized impact density for case m , $a_m = 1/4$ its initial weight, $L_d(x)$ the debris-drift likelihood, and $L_s(x)$ the probability searches would miss wreckage at x . The equations below update both locations and case probabilities. Set $L_d = 1$ for flight-only results and $L_s = 1$ before searches. This follows Bayesian conditioning and model averaging (Hoeting et al., 1999, p.383).

Drift uses nine recovered objects under a declared GLORYS12/WAVERYS current, wave and object-response model. Drift maps condition each case on available model support before assigning the equal weights; omitted support is reported. Search likelihoods preserve footprint holes. ATSB detection values of 50%, 80% and 94% test uncertainty; 94% is an approximate reference derived from ATSB (2017), Figure 73, p.96. An approximate 2018 outline is tested separately at 60%, with shared or independent missed detections. Recent unlocalized coverage is unassessed.

Across the displayed aerodynamic models and numerical repeats, the drift-plus-ATSB94 mixture places 70–92% north of 35°S. The graphics show 50/90/99% probability contours, with 25 km display smoothing. Original weights determine probabilities. This range is sensitivity, not a confidence interval. Residual weighted diversity is equivalent to 2–76 trajectory families; repeated descendants are not independent evidence. Physical calibration and numerical convergence remain incomplete, so precise hotspots are provisional. Aerodynamic alternatives remain separate. Future calibrated hydroacoustic likelihoods could similarly update locations and case weights; none is applied here.

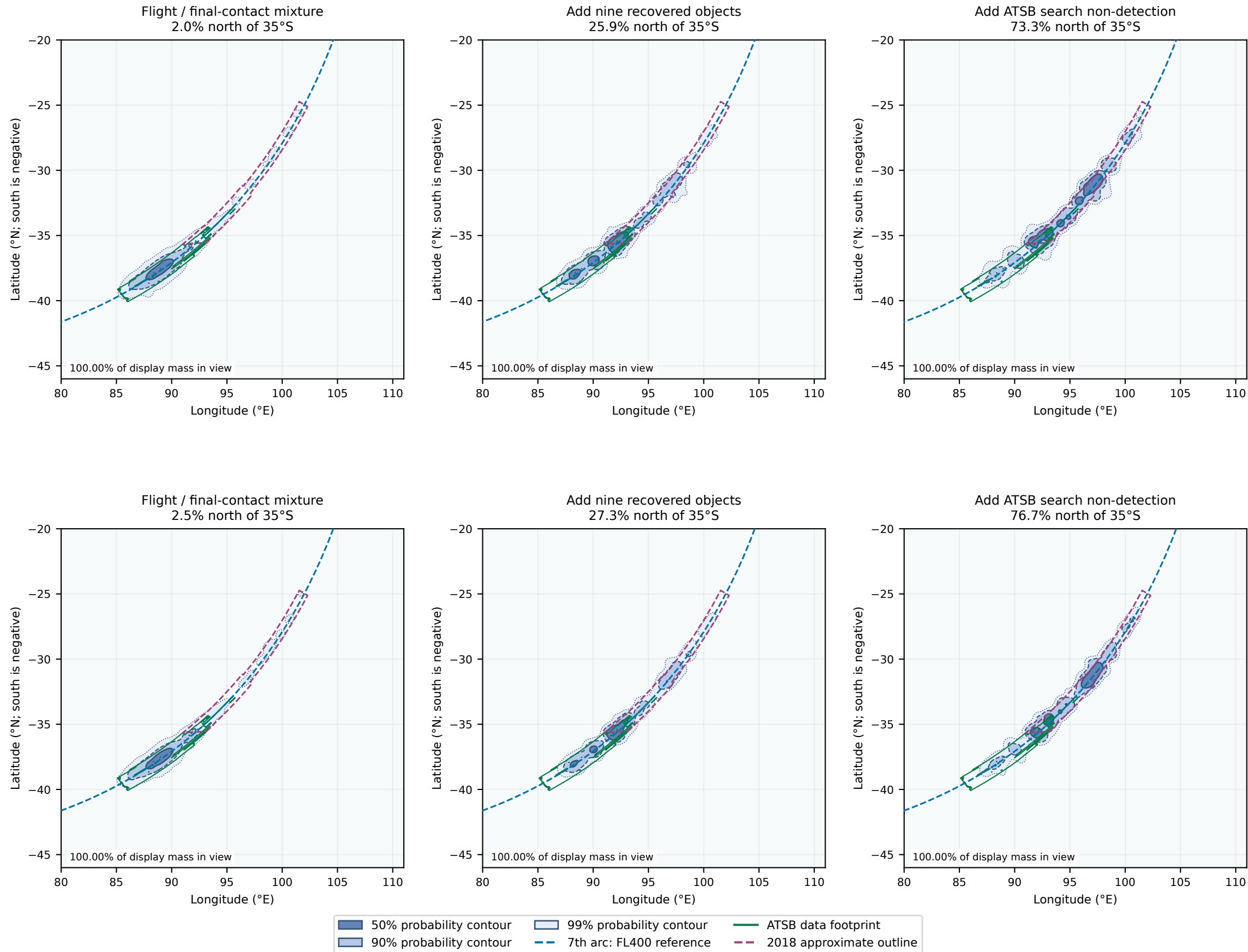
$$Z_m = \int q_m(x)L_d(x)L_s(x) \, dx$$

$$a'_m = \frac{a_m Z_m}{\sum_k a_k Z_k}$$

$$p_{\text{res}}(x) = \frac{\sum_m a_m q_m(x)L_d(x)L_s(x)}{\sum_k a_k Z_k}$$

How the evidence changes the combined impact distribution

Terminal run A · Cowling drift repeat A · Top: lower drag; bottom: higher drag



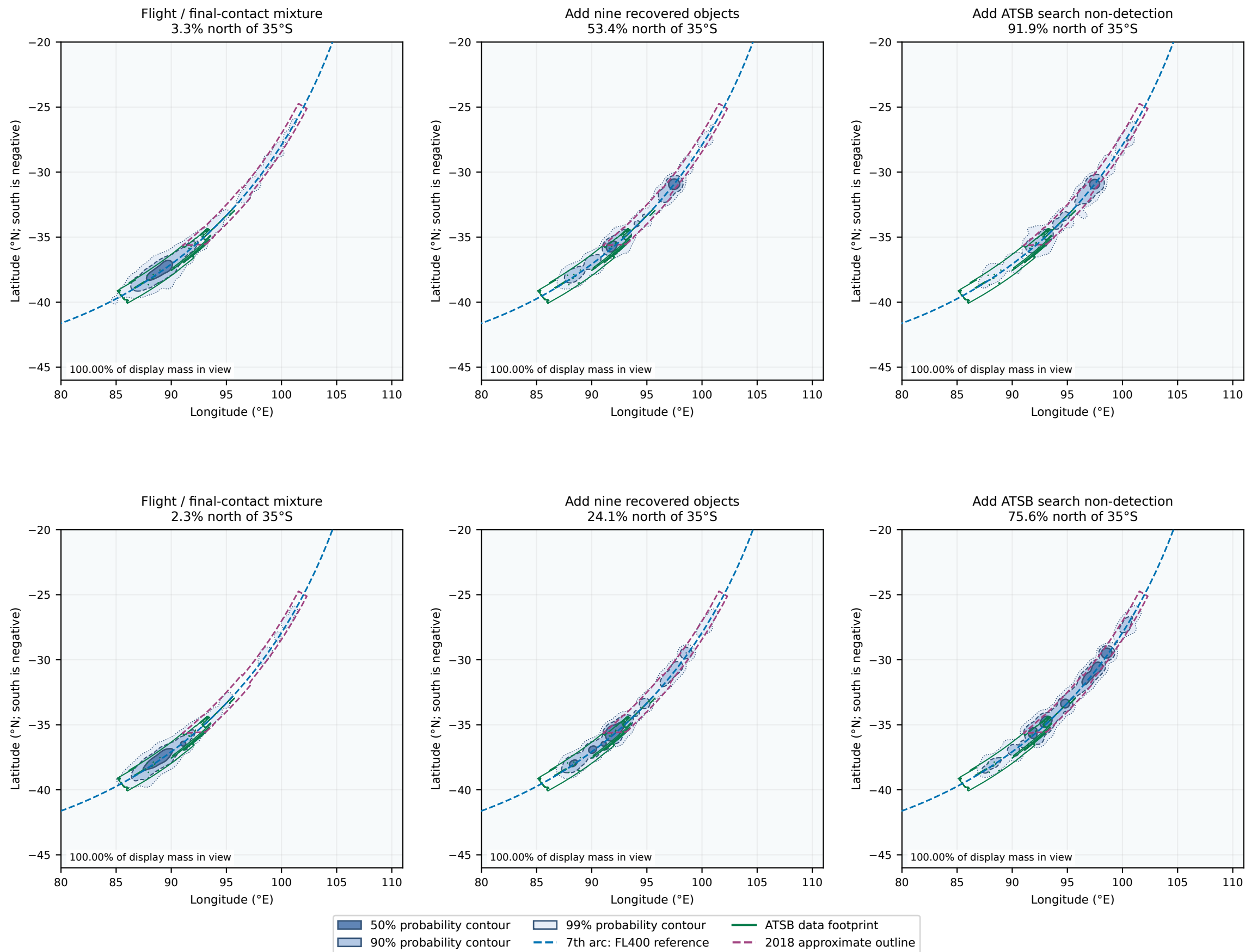
Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.

5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.

Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

How the evidence changes the combined impact distribution

Terminal run B · Cowling drift repeat A · Top: lower drag; bottom: higher drag



Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.

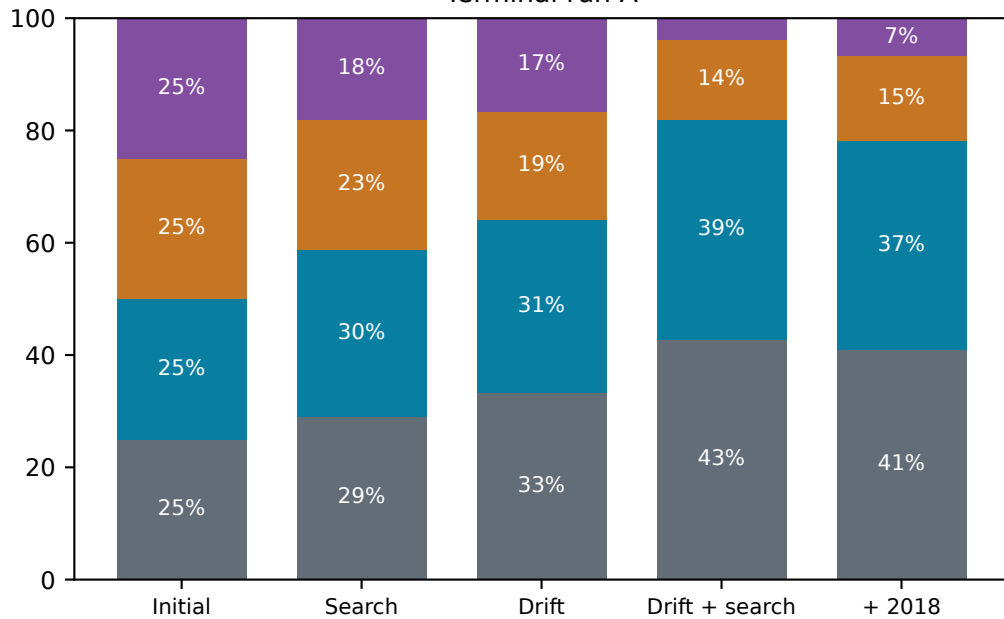
5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.

Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

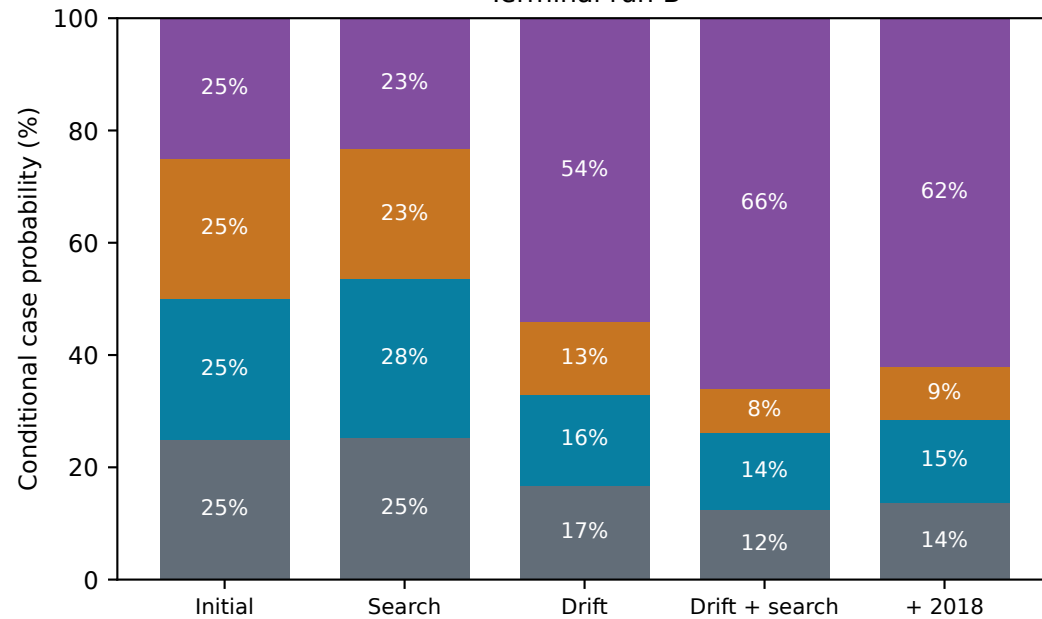
How drift and unsuccessful searches update the four 00:19 cases

Lower-drag terminal model (OpenAP 26) Equal 25% starting weights after each separate SATCOM fit

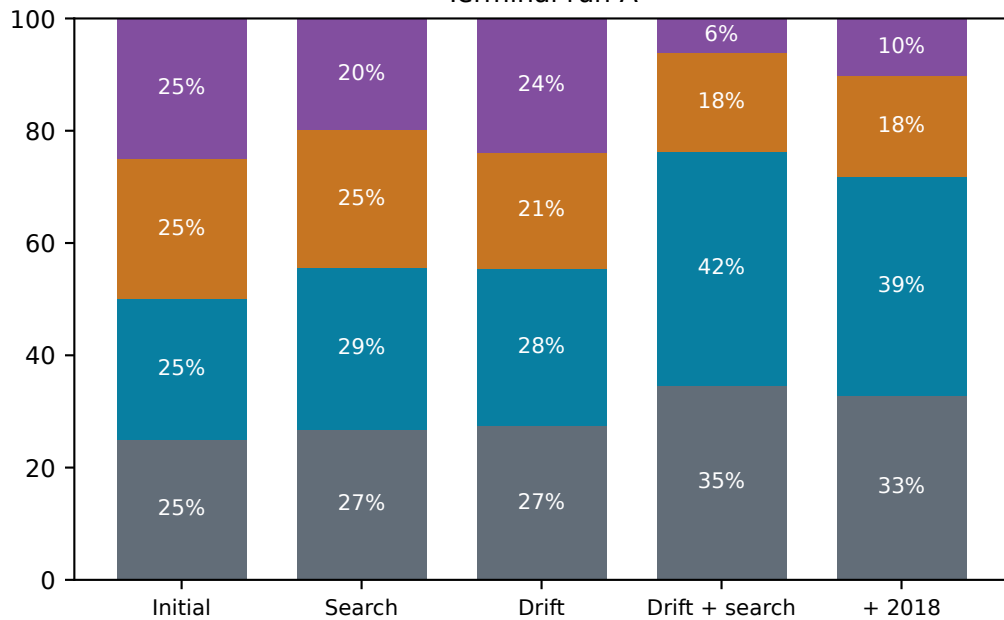
Terminal run A



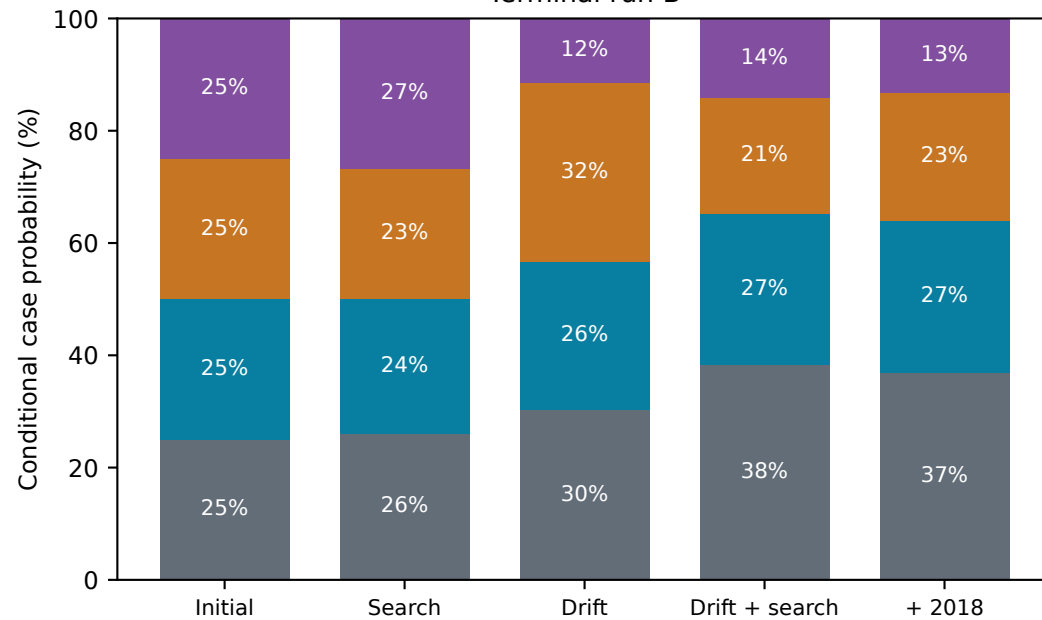
Terminal run B



Higher-drag terminal model (OpenAP 2020)
Terminal run A



Higher-drag terminal model (OpenAP 2020)
Terminal run B



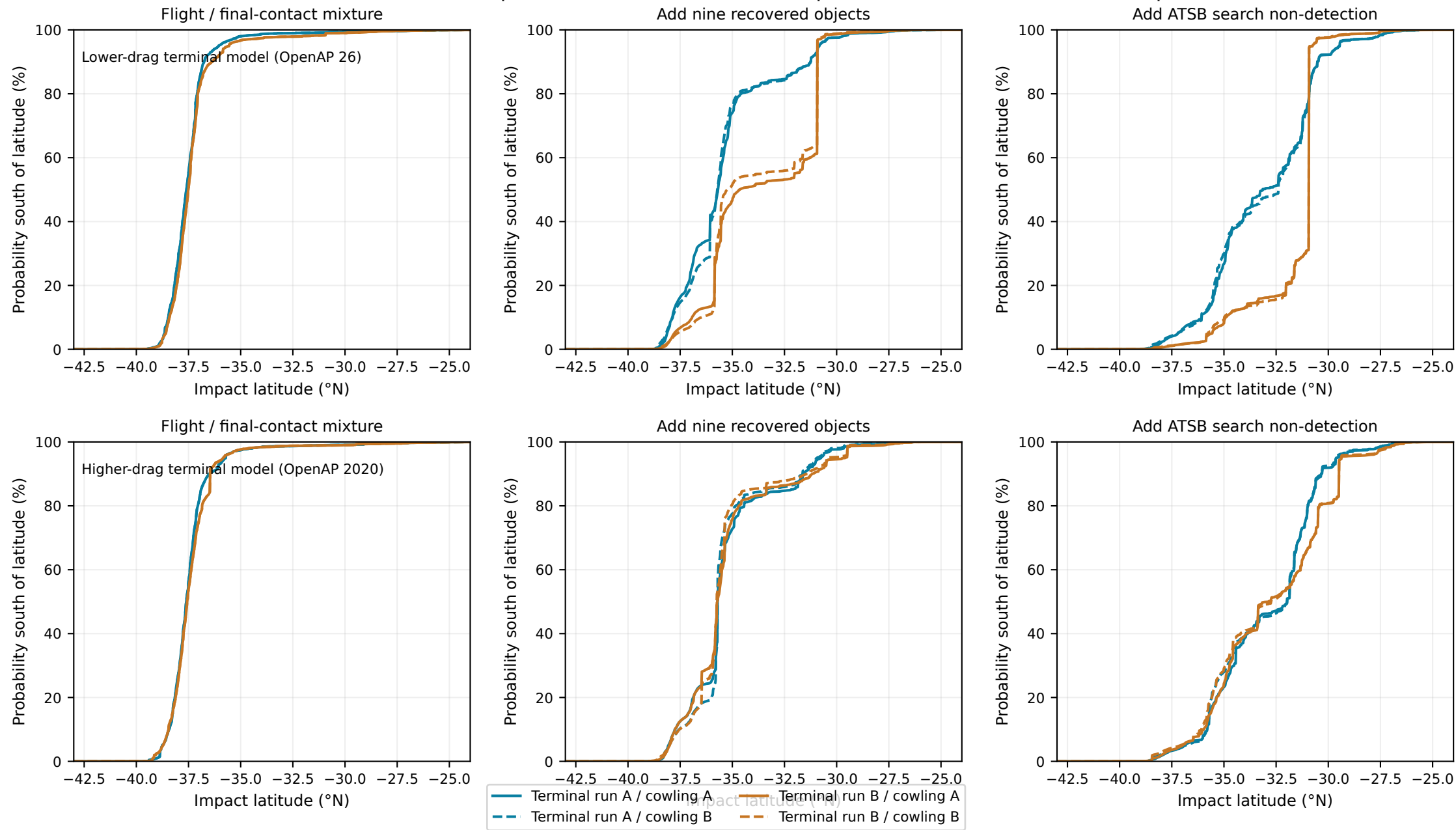
Neither BFO R600 BFO R1200 BFO Both BFOs

Search = ATSB 94% reference. +2018 = approximate outline, 60% effective detection and shared misses.

Cowling estimate A throughout. Different aerodynamic models and independent terminal runs are shown separately; hydroacoustics is absent.

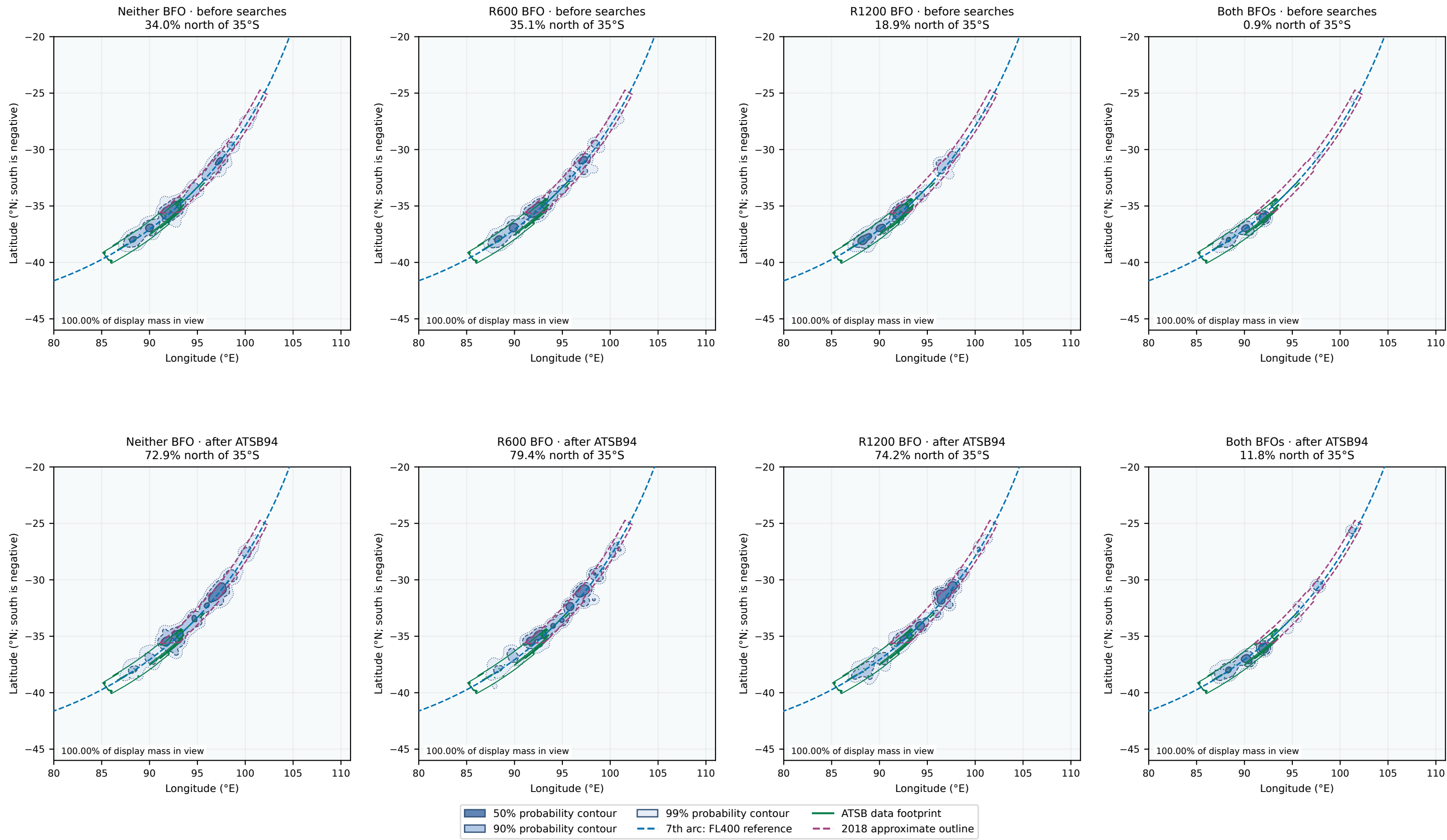
Does the combined impact distribution reproduce?

Unsmoothed cumulative probabilities; the same cruise pool is shared across terminal repeats



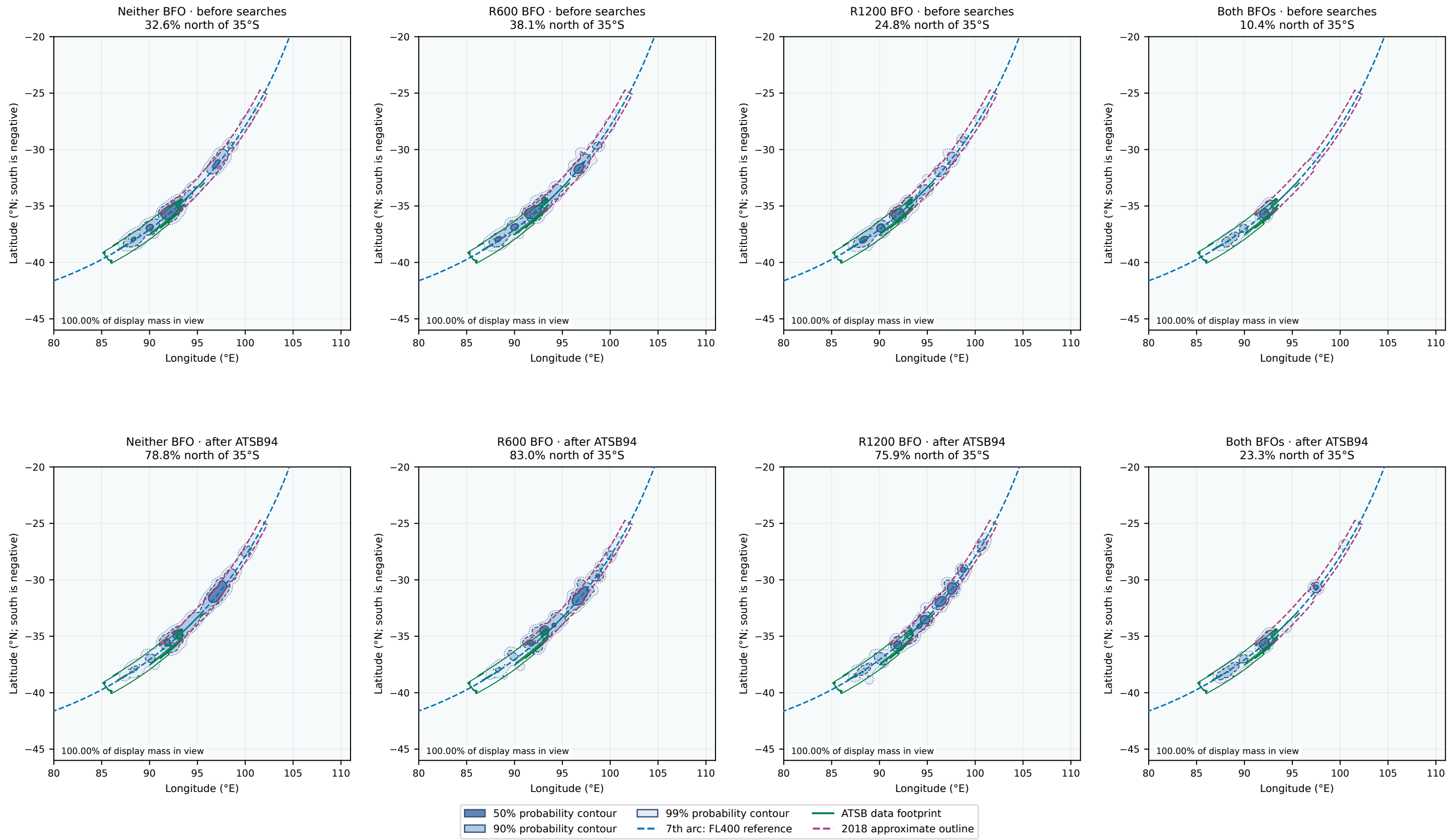
Different physical models are separate rows. Run differences measure numerical variability; these curves are not confidence bands.

Four conditional impact PDFs and their search residuals
Nine recovered objects · Lower-drag terminal model (OpenAP 26) · Terminal run A · cowling drift repeat A



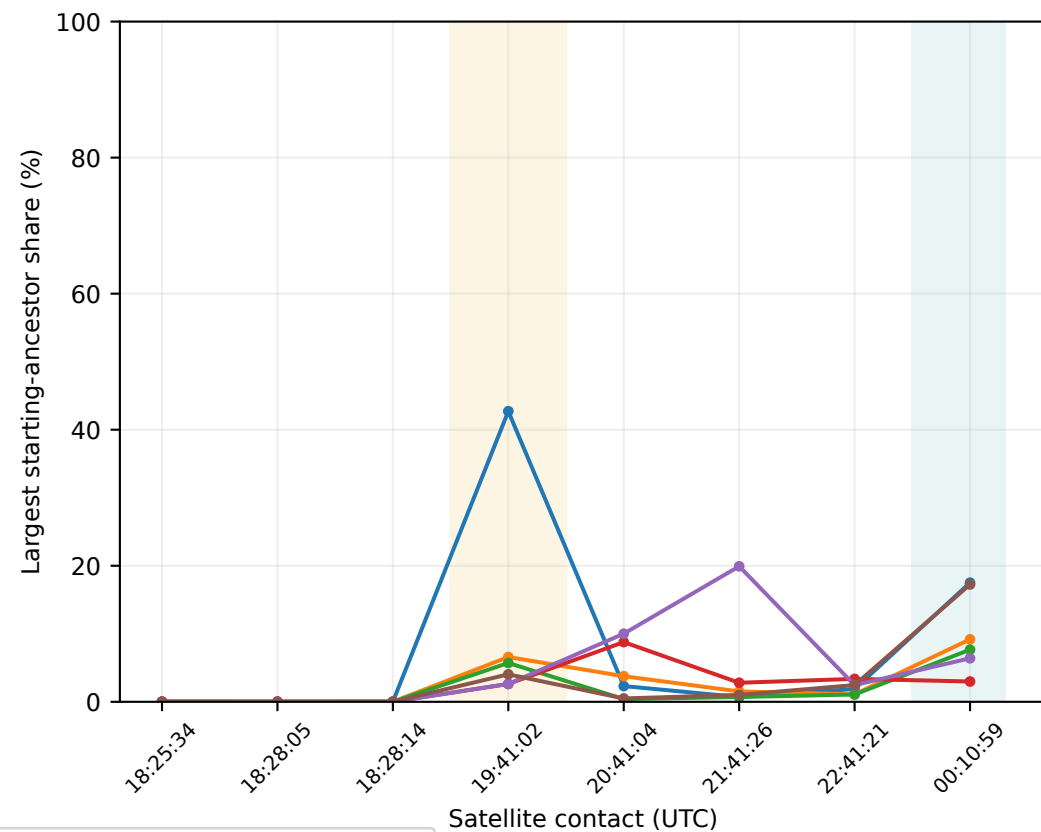
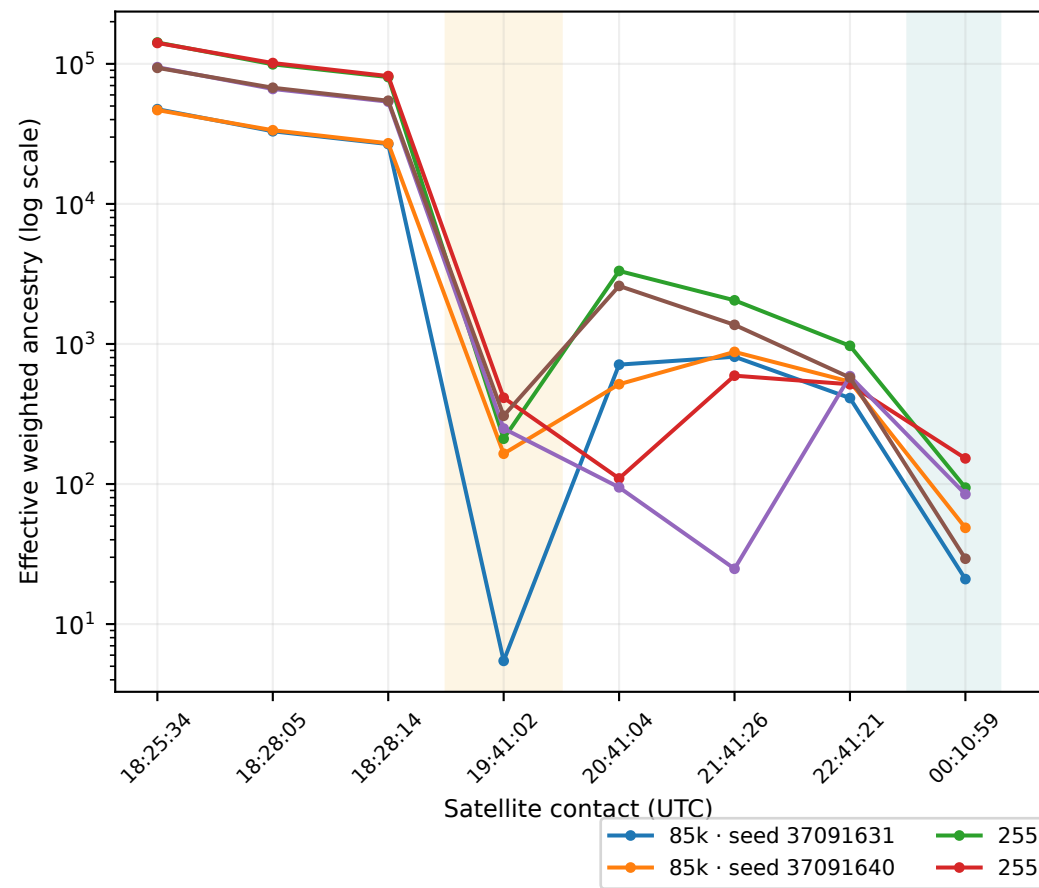
Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.
5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.
Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

Four conditional impact PDFs and their search residuals
Nine recovered objects · Higher-drag terminal model (OpenAP 2020) · Terminal run A · cowlng drift repeat A



Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.
5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.
Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

Why a million starting candidates are not a million independent posterior samples

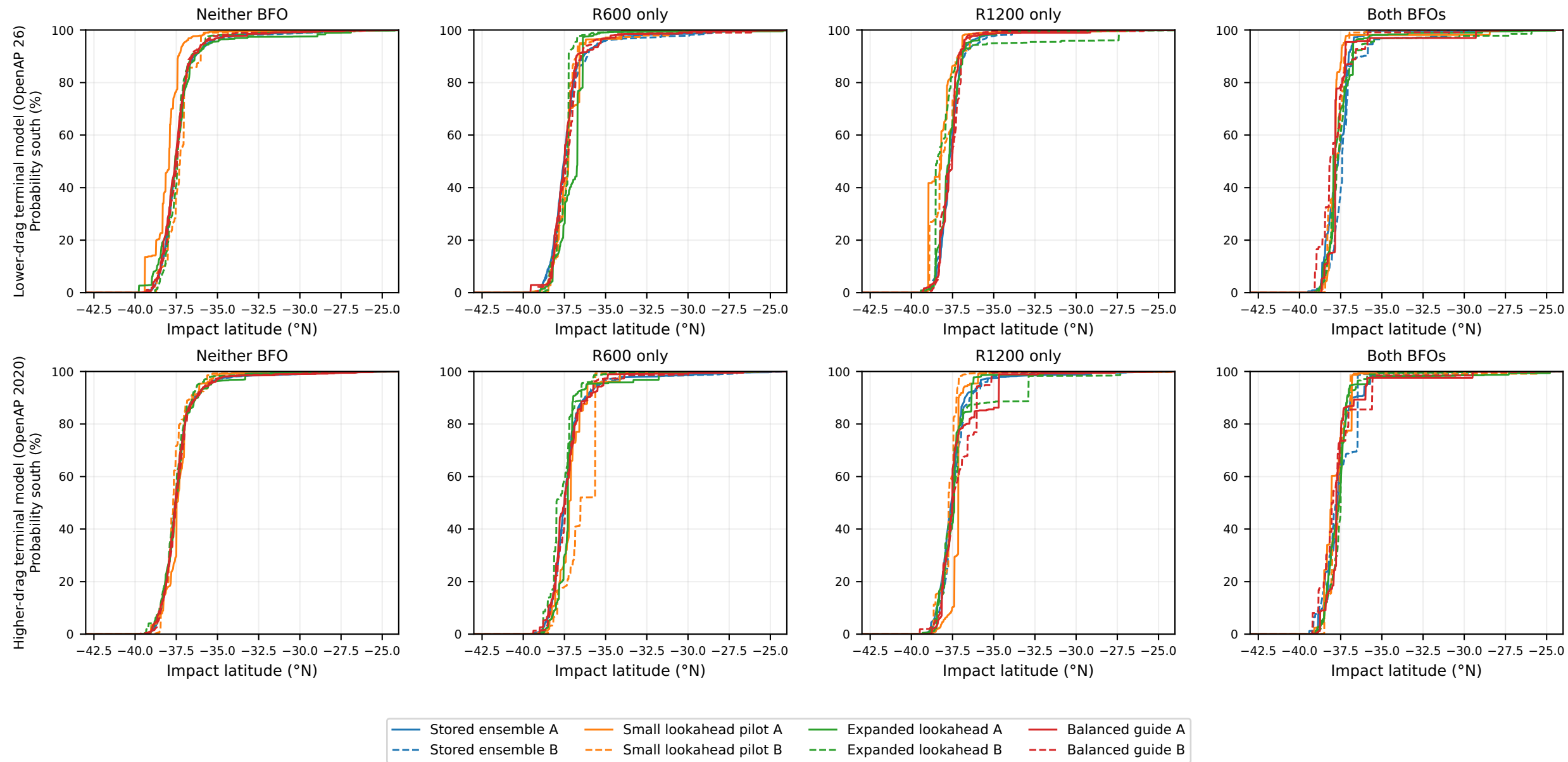


Amber: first long observation gap. Blue: final contact and fuel conditioning. Six original cruise runs; no new simulation.

Effective ancestry = $1/\sum W_r^2$, where W_r sums descendant weights from starting ancestor r .

This diagnoses weight concentration and shared ancestry; it is not a calibrated independent sample count or a convergence threshold.

Independent numerical repeats under unchanged physical assumptions



All retain corrected R600 BTO and computed impacts after 00:19:37.443. Drift and searches are absent in this sampler comparison.

Different sample counts and measured costs are recorded in the accompanying table. Pilot agreement alone does not establish accuracy.